Questions — Long-Term Innovation Strategy

Analytical Questions

1. Explain how technology roadmaps function as hierarchical planning models in Active Inference. How does the hierarchical structure (vision, milestones, actions) correspond to deep temporal models with different time horizons and uncertainty levels?

2. Analyze the three-horizon innovation portfolio framework through Active Inference. How does each horizon correspond to a different level of model confidence, and why must organizations deliberately protect Horizon 3 investments from Horizon 1 pressures?

3. James March demonstrated that organizations naturally drift toward exploitation at the expense of exploration. Explain this drift in Active Inference terms, and describe three organizational mechanisms that can counteract it.

4. How does Robert Cooper's Stage-Gate model function as a sequential planning process with evidence-based decision points? What is the Active Inference principle that should govern the precision of evidence required at each gate?

5. Compare the planning approaches of Alphabet's X Development ("monkey-first") and traditional corporate R&D (build incrementally from current capabilities). What are the Active Inference trade-offs of each approach? When is each more appropriate?

6. Explain why "no plan survives contact with reality" in Active Inference terms. What is the relationship between plan quality and plan adaptability? Can a plan be too detailed?

7. Analyze the pharmaceutical industry's innovation pipeline as a planning system under deep uncertainty. How does the portfolio of molecules at different development stages manage the risk of individual project failure?

8. How does the concept of "commander's intent" from military strategy apply to innovation planning? What is the Active Inference rationale for specifying goals and constraints rather than detailed action sequences?

9. Compare Nokia's failure to adapt to smartphones with Microsoft's successful pivot to cloud-first strategy. What planning system characteristics enabled Microsoft's adaptation and prevented Nokia's?

10. Explain how the COVID-19 vaccine development compressed the pharmaceutical planning pipeline without eliminating essential evidence-generating steps. What changed in the planning model, and what remained constant?

Applied Questions

1. Design a technology roadmap for your invention covering 1-year, 3-year, and 5-year horizons. For each milestone, specify the achievement, the uncertainty level, the evidence that would confirm you are on track, and the conditions under which you would revise the plan.

2. Classify your invention's activities across the three innovation horizons. Assess whether your portfolio is balanced and propose adjustments. What Horizon 3 exploration are you currently neglecting?

3. Identify the "monkey" in your invention's roadmap — the hardest, most uncertain challenge. Design an experiment or test that would generate evidence about this challenge before you invest heavily in easier components of the plan.

4. Design a four-stage innovation pipeline for your invention with evidence-appropriate gate criteria. Ensure that early gates are permissive enough to allow promising ideas to develop and late gates are demanding enough to prevent wasteful launches.

5. Your industry is experiencing a strategic inflection point (a fundamental change in the competitive landscape). Design an adaptive planning response: what signals would you monitor, what plan revisions would each signal trigger, and how would you maintain organizational alignment during the transition?

6. You have limited resources and must choose between three strategic options: (a) optimize your current product for your current market (exploitation), (b) extend your product to an adjacent market (moderate exploration), or (c) invest in a fundamentally

new technology that could create a new market (radical exploration). How do you analyze this decision using Active Inference's expected free energy framework?

7. Design a "commander's intent" document for your invention that specifies the non-negotiable vision and principles while leaving specific plans flexible. How would this document guide decision-making when unexpected circumstances arise?

8. Your invention has been successful in its initial market, and you are planning expansion. Design a multi-phase expansion plan that sequences market entries to maximize learning from each phase. How does evidence from Phase 1 inform the design of Phase 2?

9. Create a strategic review system for your invention: define the frequency, participants, key questions, and outputs of routine reviews, milestone reviews, and emergency reviews. What evidence thresholds trigger an emergency review?

10. Reflecting on the entire Innovation Ecosystems section (Modules 01-08), design an integrated innovation strategy for your invention that synthesizes ecosystem analysis, stakeholder engagement, market sensing, strategic thinking, launch planning, organizational learning, communication strategy, and long-term planning into a coherent, adaptive plan.